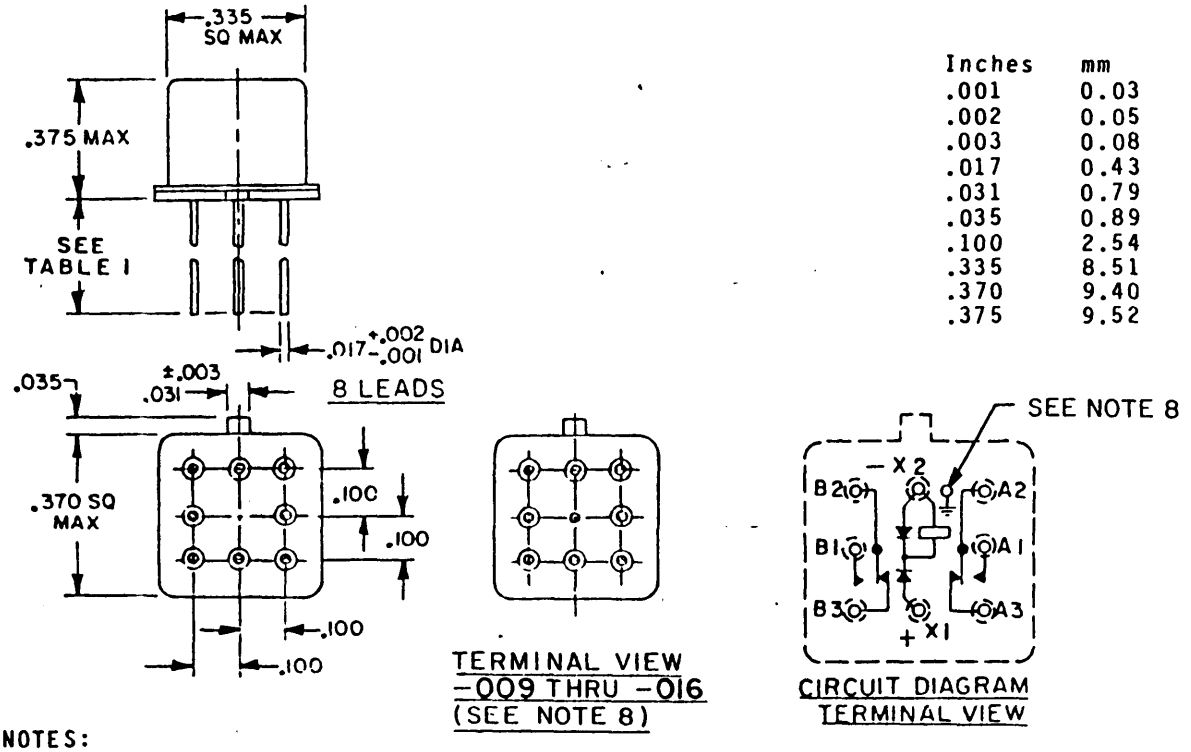
MILITARY SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, ESTABLISHED RELIABILITY, DPDT, LOW LEVEL TO 1.0 AMPERE WITH INTERNAL DIODES FOR COIL TRANSIENT SUPPRESSION AND POLARITY REVERSAL PROTECTION TERMINALS 0.100-INCH GRID PATTERN (SENSITIVE 60 MILLIWATTS COIL OPERATE POWER AT 25°C), INCLUDING INSPECTIONS OF APPENDIXES A AND B

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-R-39016.



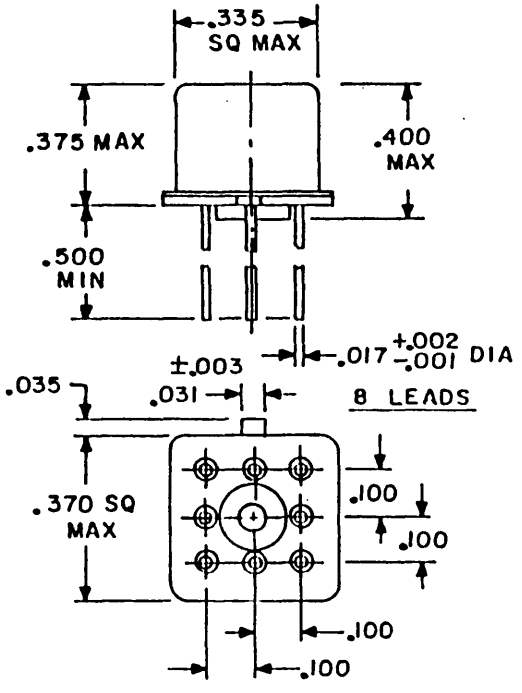
Inches	mm
.001	0.03
.002	0.05
.003	0.08
.017	0.43
.031	0.79
.035	0.89
.100	2.54
.335	8.51
.370	9.40
.375	9.52

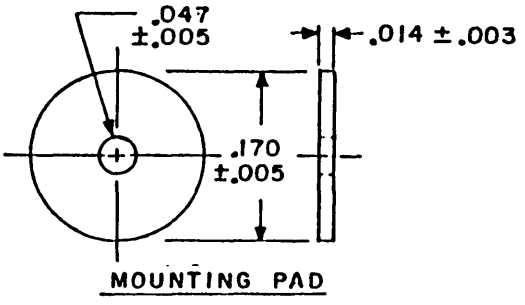
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Terminal numbers shown above for reference only. Numbers do not appear on relay.
5. Relays shall have a plus (+) sign placed on the circuit diagram as shown.
6. Coil symbol optional in accordance with MIL-STD-1285.
7. Circuit diagram shown on part is the terminal view.
8. The grounding pin shown is a noninsulated case grounding pin applicable to -009 through -016 only.

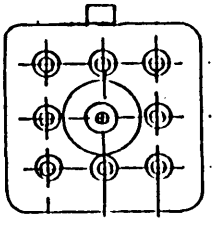
FIGURE 1. Dimensions and configuration.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

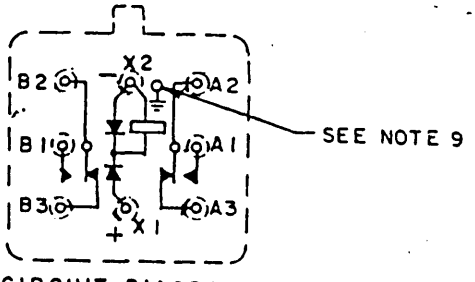
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TERMINAL VIEW
APPLICABLE TO
-033 THRU -040 ONLY
(SEE NOTE 9)



CIRCUIT DIAGRAM
TERMINAL VIEW

Inches	mm
.001	0.03
.002	0.05
.003	0.08
.005	0.13
.014	0.36
.017	0.43
.031	0.79
.035	0.89
.047	1.19
.100	2.54
.170	4.32
.335	8.51
.370	9.40
.375	9.52
.400	10.16
.500	12.70

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Terminal numbers shown above for reference only. Numbers do not appear on relay.
5. Relays shall have a plus (+) sign placed on the circuit diagram as shown.
6. Coil symbol optional in accordance with MIL-STD-1285.
7. Circuit diagram shown on part is the terminal view.
8. Mounting pad shall be a polyester film in accordance with MIL-I-631, type G, class I.
9. The grounding pin shown is a noninsulated case grounding pin applicable to -033 through -040 only.

FIGURE 2. Dimensions and configuration (relay with mounting pad).

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REQUIREMENTS:

Cleaning and small particle inspection: Applicable.

CONTACT DATA:

Load ratings:

High level (relay case grounded):

Resistive:

1.0 ampere at 28 V dc.

500 milliamperes at 115 V ac 400 Hz case not grounded.

250 milliamperes at 115 V ac 60 Hz case not grounded.

100 milliamperes at 115 V ac 60 and 400 Hz case grounded.

Inductive: 0.2 ampere at 28 V dc with 0.32 henry inductance.

Lamp: 0.10 ampere at 28 V dc.

Low level: 10 to 50 µA at 10 to 50 mV dc or peak ac.

Intermediate current: Applicable.

Contact resistance or voltage drop:

Initial: 0.10 ohm maximum (0.110 maximum with spacer pad).

High level:

During life: Not more than 5 percent of circuit voltage.

After life: 0.20 ohm maximum (0.210 maximum with spacer pad).

Low level:

During life: 100 ohms maximum.

After life: 0.15 ohm maximum (0.160 maximum with spacer pad).

Intermediate current:

During: 3 ohms maximum.

After: 0.20 ohm maximum (0.210 maximum with spacer pad).

Contact bounce: 1.5 milliseconds (ms) maximum (applicable for after high level rated load life and intermediate current tests).

Contact stabilization time: 2.5 milliseconds maximum.

Overload (high level only): Two times rated current.

COIL DATA: See table I.

Operate time: 4.0 ms maximum over temperature range with rated coil voltage.

Release time: 7.5 ms maximum over temperature range from rated coil voltage.

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ELECTRICAL DATA:

Insulation resistance 1/: 10,000 megohms minimum at 500 V dc, except the resistance between coil and case at high temperature shall be 1,000 megohms minimum.

Dielectric withstanding voltage: 1/

	Sea level V rms (60 Hz)	Altitude V rms (60 Hz)
Between case, frame, or enclosure and all contacts both in the energized and deenergized positions	500	125 All terminals to case
Between case, frame, or enclosure and coil	500	
Between all contacts and coil	500	
Between open contacts in the energized and deenergized positions	500	
Between contact poles	500	
Between coils of dual coil relays	---	

DIODE CHARACTERISTICS:

Coil transient suppression: Applicable.

Diode breakdown and block integrity (delete coil resistance and substitute the test in all inspection tables of MIL-R-39016):

With applicable voltage applied to the relay coil circuit in the reverse direction, monitor leakage current with dc microammeter or oscilloscope or qualifying activity approved test equipmemt. Leakage current shall not exceed the specified value.

Maximum negative transient: 1.0 volt.

Breakdown voltage: 100 V dc minimum at 10 microamperes (μA).

Maximum leakage current: 1 μA at 50 V dc.

Peak inverse voltage: 100 volts minimum.

Semiconductor in-process screening: Applicable, visual inspection of semiconductors shall be in accordance with MIL-STD-750, method 2073 or 2074.

ENVIRONMENTAL DATA:

Temperature range: -65°C to +125°C.

Vibration (sinusoidal): MIL-STD-202, method 204. Contact chatter shall not exceed 10 microseconds maximum for closed contacts, and 1 microsecond maximum closure for open contacts.

1/ Insulation resistance and dielectric withstanding voltage tests must always precede all other specified electrical measurements. Connect all coil terminals together to avoid damage to diodes.

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TABLE I. Dash numbers and characteristics. 1/

Dash number 2/			Coil voltages (V dc) 4/		At +25°C						Over temperature range		
					Coil resistance (ref. only) Ohms 5/	Coil circuit current 5/ (mA)		Specified pickup value (voltage) (V dc)	Specified hold value (voltage) (V dc)	Specified drop-out value (voltage) (V dc)	Specified pickup value (voltage) (V dc)	Specified hold value (voltage) (V dc)	Specified dropout value (voltage) (V dc)
Lead length .500 Min	Lead length .187 ±.010	Mounting pads (fig. 2) 3/	Rated	Max		Max	Min						
001	017	025	5.0	7.0	64	78.1	56.8	2.9	2.2	0.8	3.7	2.6	0.7
002	018	026	6.0	10.0	125	48.9	36.3	4.0	2.5	0.9	4.8	3.0	0.8
003	019	027	9.0	15.0	400	23.6	18.1	6.1	3.6	1.1	8.0	4.5	0.9
004	020	028	12.0	20.0	800	16.0	12.5	7.8	4.6	1.3	11.0	5.8	1.0
005	021	029	18.0	30.0	1,600	12.2	9.6	11.3	7.0	1.5	14.5	9.0	1.1
006	022	030	26.5	40.0	3,200	9.0	7.2	15.2	10.8	1.7	19.0	13.0	1.3
007	023	031	36.0	57.0	6,500	6.1	4.9	21.7	14.7	2.3	27.2	19.0	1.7
008	024	032	48.0	75.0	11,000	4.8	3.9	27.8	19.8	2.8	34.8	26.0	2.0
009 6/	---	033 6/	5.0	7.0	64	78.1	56.8	2.9	2.2	0.8	3.7	2.6	0.7
010 6/	---	034 6/	6.0	10.0	125	48.9	36.3	4.0	2.5	0.9	4.8	3.0	0.8
011 6/	---	035 6/	9.0	15.0	400	23.6	18.1	6.1	3.6	1.1	8.0	4.5	0.9
012 6/	---	036 6/	12.0	20.0	800	16.0	12.5	7.8	4.6	1.3	11.0	5.8	1.0
013 6/	---	037 6/	18.0	30.0	1,600	12.2	9.6	11.3	7.0	1.5	14.5	9.0	1.1
014 6/	---	038 6/	26.5	40.0	3,200	9.0	7.2	15.2	10.8	1.7	19.0	13.0	1.3
015 6/	---	039 6/	36.0	57.0	6,500	6.1	4.9	21.7	14.7	2.3	27.2	19.0	1.7
016 6/	---	040 6/	48.0	75.0	11,000	4.8	3.9	27.8	19.8	2.8	34.8	26.0	2.0

1/ Each relay possesses high level and low level capabilities. However, relays previously tested or used above 10 mA resistive at 6 V dc maximum or peak ac open circuits not recommended for subsequent use in low level applications.

2/ The suffix letter M, P, or R to designate the applicable failure rate level shall be added to the applicable listed dash number. Failure rate level (percent per 10,000 cycles): M, 1.0; P, 0.1; R, 0.01. Example, 001M - - - - - 040R.

3/ Relays supplied with mounting pads (-025 through -040) shall have the pad rigidly attached.

4/ Caution: The use of any coil voltage less than the rated coil voltage will compromise the operation of the relay.

5/ Coil resistance not directly measurable at relay terminals. When rated voltage is applied to the coil terminals, the coil circuit current must be within the limits shown. Measure at 25°C at nominal voltage for 5 seconds, maximum.

6/ Relays are supplied with a case grounding pin welded to the header (see figures 1 and 2).

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Vibration (random): MIL-STD-202, method 214, test condition IG. Contact chatter shall not exceed 10 microseconds maximum for closed contacts, and 1 microsecond maximum closure for open contacts. Applicable to qualification, group C, and group A inspection testing.

Shock (specified pulse): MIL-STD-202, method 213, test condition B (75 g's). Contact chatter shall not exceed 10 microseconds maximum for closed contacts, and 1 microsecond maximum closure for open contacts.

Magnetic interference: Applicable.

Resistance to soldering heat: Applicable.

Acceleration: Applicable.

Salt atmosphere (corrosion): In accordance with MIL-STD-750, method 1041.

Particle impact noise detection: Applicable.

PHYSICAL DATA:

Terminals:

Terminal strength: 1 pound pull minimum.

Terminal twist test: As specified in MIL-R-39016.

Dimensions and configuration: See figures 1 and 2.

Terminations: See figure 1 and table I.

Weight: 4.3 grams (0.15 ounce) maximum.

Solderability: Applicable.

Seal: Hermetic, adjunct sealant shall not be used.

Minimum marking: Military PIN, "J" with the date code (example J8830), circuit diagram, and manufacturer's name or source code.

LIFE TEST REQUIREMENTS:

High level: 100,000 cycles per relay.

Low level: 100,000 cycles plus 900,000 cycles mechanical life.

Part or Identifying Number (PIN): M39016/643- (dash number from table I and suffix letter designating failure rate level).

QUALIFICATION INSPECTION:

Qualification inspection and sample size: See table II.

Failure rate qualification: The manufacturer shall qualify his products to failure rate level "M" initially.

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TABLE II. Qualification inspection and sample size. 1/

Single submission	Group submission	
33 units plus 1 open unit for level M at C = 0 2/ Qualification inspection as applicable	M39016/643-006	33 units plus 1 open unit for level M at C = 0 2/ Qualification inspection as applicable
	M39016/643-001 M39016/643-002 M39016/643-003 M39016/643-004 M39016/643-005 M39016/643-007 M39016/643-008	2 units each PIN Qualification inspection, group I group II
	M39016/643-014	1 unit terminal strength and solderability
	M39016/643-030	2 units each PIN 3/

1/ For retention of qualification or extension of qualification to lower failure rate levels, all life test data accumulated on MIL-R-39016/43 may be used in addition to MIL-R-39016/643. Prior to performance of retention of qualification testing, the relay manufacturer shall preselect the sampling plan.

2/ The number of units required for qualification testing shall be increased as required in group V, table II, MIL-R-39016, if the contractor elects to test the number of units permitting one or more failures. Prior to performance of qualification testing, the relay manufacturer shall preselect the sampling plan.

3/ Qualification inspection of relays supplied with spreader pads shall be tested as specified in the qualification inspection table of MIL-R-39016 in the order shown below:

Before installation of the pad:

Vibration (sinusoidal) test duration shall be 10 minutes, vibration (random), particle impact noise detection, screening.

After installation of the pad:

Insulation resistance, dielectric withstanding voltage, static contact resistance, specified pickup, hold, and dropout values (voltages), coil resistance, operate and release time, contact dynamic characteristics, coil transient suppression (when specified), solderability, seal, visual and mechanical inspection (external).

Group A inspection: Group A inspection for relays supplied with spreader pads shall be as specified below:

Before installation of the pad: Perform subgroups 1 and 2 of the group A inspection table.

After installation of the pad: Perform subgroups 3 and 4 of the group A inspection table.

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Qualification inspection (reduced testing) and sample size: If the relays produced for MIL-R-39016/643 are similar in construction and design, except for the additional inspections specified herein, to the relays produced for MIL-R-39016/43, then reduced testing may be performed concurrent with or subsequent to successful qualification of MIL-R-39016/43 at the "M" failure rate level. See table III.

TABLE III. Qualification inspection (reduced testing).

Inspection
2 units each coil voltage, group A inspection table.
1 unsealed sample unit for internal inspection.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Navy - AS
Air Force - 99
DLA - ES
NSA - NS

User activities:
Army - AR
Navy - MC, OS, SH
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5945-0766-16)

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